Customer Behavior and Marketing Analysis: Predicting Customer Income Using Machine Learning

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***Abstract*—**The aim of this research is to predict the earnings of the customers by analyzing the demographic data and shopping behavior of the customers using various machine learning techniques. The data used in this research is from Kaggle and it is called the Customer Behavior Dataset. It has 100,000 customer entries with different attributes including age, gender, education, location, loyalty level, frequency of purchasing, total spending, participation in promotions and customer satisfaction. To get the better understanding of the characteristics of the dataset, we did Exploratory Data Analysis (EDA). This phase was important to visualize the distribution of data, the behavior of customers and the relation of different variables with each other. The analysis showed a strong correlation between the expenditure and income variables, while other variables had only a small correlation with the primary target. Before developing the model, we cleaned the data set by applying label encoding to categorical features and dividing the data into two subsets, one for training and one for testing. Then we built and evaluated three different regression models: Linear Regression, Random Forest Regressor, and SGD Regressor.

The results show that the Random Forest model gives the most accurate predictions among the models examined. Its ability to detect non-linear relationships makes it better at predicting customer income than the other techniques. Linear Regression works well for individuals with lower incomes, but its accuracy tends to decline as income increases. Conversely, the SGD Regressor is prone to making more errors and yields results that lack consistency. In conclusion, the results highlight that machine learning techniques can effectively be employed to project customer income and assess behavior patterns. Among all the various models, Random Forest has demonstrated to be the most efficient with this particular dataset.

I. Introduction

In the rapidly evolving era of modern e-commerce, the challenge to remain competitive and maintain good customer relationships often leads businesses to implement new innovation or methods with technologies to increase revenue [1]. Consumers are individuals who use goods or services provided by producers for their own personal benefit or others. Every consumer shows diverse behaviors, so their responses to specific products or brands also vary [2]. By analyzing insights from customer behavior analysis, companies can optimize their marketing strategies, enhance the customer experience, and refine product offerings, which ultimately increases customer satisfaction and loyalty [3]. Customer data can provide valuable insights into purchasing patterns, customer segmentation, and revenue potential, thereby helping businesses make more informed and strategic decisions [4].

The implementation of predictive analysis into customer behaviour and relationships could help businesses or organizations to explore more beyond descriptive analysis [5]. Among the analyzed metrics, predicting income level is crucial to manage financial product offerings to customers. There are many ways to predict a customer's income without relying on credit scores [6]. With the rise of machine learning development, it enables enterprises to analyze patterns with customers behavior with multiple features, which credit scores may not capture. Based on other research, ensemble models such as Random Forest, outperforms traditional statistical baselines in predicting income as a classification or regression task based on specific features [7]. By deploying scalable algorithms, businesses can accurately map complex demographic interactions while making strategic marketing decisions.

II. Literature Review

Customer behavior analytics has gained significant attention in recent years due to the proliferation of digital transaction data. [8] demonstrated that machine learning techniques can effectively predict consumer behavior from large-scale transaction data, enabling more targeted marketing campaigns. Their work established robust frameworks for applying predictive models to consumer purchasing data.

Exploratory Data Analysis (EDA) serves as a critical preliminary step in any data science pipeline. [9] advanced EDA by proposing automated approaches leveraging

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machine learning to discover meaningful patterns and summarize data characteristics efficiently. EDA can be used in customer analytics to identify distributions, outliers and correlations that can inform subsequent modelling choices, especially when combined with modern preprocessing techniques [10].

A commonly used technique for outlier identification is the Interquartile Range (IQR), which is robust to non-normal distributions and defines boundaries at Q1 - 1.5×IQR and Q3 + 1.5×IQR. [11] developed this IQR based outlier detection framework and proved its effectiveness on various datasets. This approach is particularly appropriate for customer datasets with potentially skewed distributions of spend patterns.

Linear Regression, which assumes a linear relationship between features and target variable, is a fundamental baseline for regression problems [12]. Although it is interpretable and computationally efficient, it may not perform well if the true relationship is non-linear. [13] reviewed ensemble learning methods comprehensively and showed how multiple decision trees are developed by strategies such as Random Forest and their predictions are merged to greatly improve predictive accuracy and robustness compared to single-model methods.

Stochastic Gradient Descent SGD offers an iterative approach to optimise large scale datasets [14]. However, SGD is sensitive to feature scaling and requires careful preprocessing and hyperparameter tuning to converge [15]. Several studies have demonstrated that inappropriate scaling results in model divergence, as evidenced by our experimental results.

Hyperparameter tuning with cross-validation using GridSearchCV is a common approach to obtain the best performance of the model. [16] applied GridSearchCV to tune machine learning classifiers for a medical prediction task, showing its capability of systematically traversing the hyperparameter space. This method evaluates combinations of hyperparameters while using a k-fold cross-validation technique to estimate the generalisation performance, that reduces the risk of overfitting [17].

Correlation analysis between variables offers important insights for feature selection. [18] investigated the relationship between income, spending behaviour, and e-commerce patterns, and found strong linkages between customer income and purchase amounts in digital retail environments. [19] also showed that predictive analytics of consumer behaviour can accurately model these income-driven purchasing patterns, highlighting the importance of income as a primary factor influencing consumer decision-making.

III. Methodology

A. Dataset

This study uses the Customer Behavior Dataset from Kaggle [20]. It has 100,000 customer records and 12 features about the customers. The features of the Customer Behavior Dataset are important because they show details about the customers and their buying habits. The Customer Behavior Dataset helps us understand how the customers make purchases. We can see a summary of the features of the Customer Behavior Dataset, in Table I.

Table 1: Dataset Column Description

Column	Type	Description
id	Numeric	Unique customer ID
age	Numeric	Customer age (12-49 years)
gender	Categorical	Male/Female
income	Numeric	Customer income (**target variable**)
education	Categorical	Bachelor, Masters, HighSchool, College
region	Categorical	East, West, South, North
loyalty_status	Categorical	Gold, Silver, Regular
purchase_frequency	Categorical	Frequent, Occasional, Rare
purchase_amount	Numeric	Total purchase amount
product_category	Categorical	Books, Clothing, Food, Electronics, Home, Beauty, Health
promotion_usage	Numeric (binary)	0 = Did not use promo, 1 = Used promo
satisfaction_score	Numeric	Customer satisfaction score (0–10)

The main goal of this study is to predict customer income, focusing on income as the variable. This project uses a mix of demographic and behavioral features to make predictions on customer income.

B. *Data Wrangling*

1) Initial Inspection

The dataset was looked at to see if it was good and complete. We found that the data was already in shape. There were no missing values or duplicate rows in the 12 features of the dataset. So we did not have to fix any values or remove any duplicate rows in the dataset before moved on to the next part.

2) Removal of Irrelevant Columns

The id attribute was taken out of the dataset because it only serves as an identifier, for each record. It does not really help with analysis or prediction. So this column was removed. The dataset now has 11 features that are useful for further processing. These features are relevant and will be used for analysis and prediction of the dataset.

3) Outlier Handling

The Interquartile Range (IQR) approach was used to identify outliers for each attribute. The Interquartile Range formula was used to determine the lower and upper bounds for each characteristic. Records that had values outside of these ranges were eliminated from the dataset because they were deemed outliers. This step improved the analysis's dependability by reducing the influence of values. The accuracy of the results was increased by identifying outliers using the Interquartile Range technique.

C. *Exploratory Data Analysis (EDA)*

1) Descriptive Statistics

Descriptive statistics were applied to summarize the numeric features, providing a better understanding of the data distribution within the dataset.

	age	income	purchase_amount	promotion_usage	satisfaction_score
count	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000
mean	30.003260	27516.269880	9634.790840	0.300800	5.009650
std	4.480535	12996.782587	4799.339449	0.458608	1.038714
min	12.000000	5000.000000	1118.000000	0.000000	0.000000
25%	27.000000	16271.750000	5583.000000	0.000000	4.000000
50%	30.000000	27584.500000	9452.000000	0.000000	5.000000
75%	33.000000	38747.250000	13350.000000	1.000000	6.000000
max	49.000000	50000.000000	26204.000000	1.000000	10.000000

Fig. 1: Descriptive Statistics Table

Figure 1 provides descriptive statistics to give a clearer picture of the customer base. The average age of the customers is 30 years, indicating a relatively young demographic. Income levels vary greatly, ranging from 5,000 to 50,000, with an average of 27,516. Data also show that only 30% of customers took advantage of promotional offers. The average satisfaction score is 5 on a 10-point scale. This means that customer satisfaction is generally moderate and not particularly high or low.

2) Numerical Data Distribution

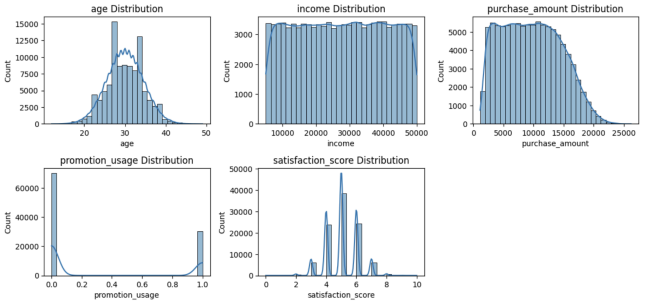


Fig. 2: Numerical Distribution Histogram

Figure 2 shows that the age of customers is normally distributed, with most people being between 27 and 33 years old. The income of customers is also spread out in an even way from 5,000 to 50,000 which means it has customers with all kinds of different incomes. The amount that people spend when they make a purchase is a bit different because some customers spend a lot more than others even though most people spend about the amount. The score that shows how satisfied customers are is close to a normal distribution with most scores being between 4 and 6 which means that customers are generally pretty satisfied with what they get.

3) Categorical Data Distribution

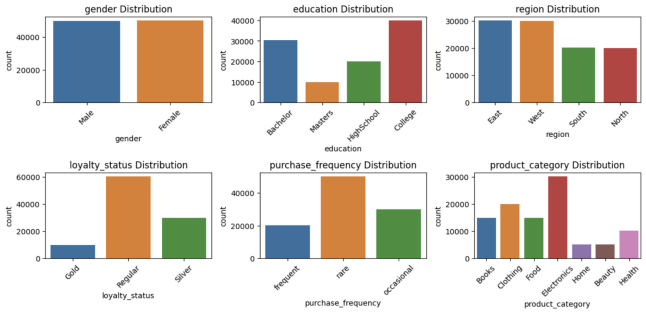


Fig. 3: Categorical Distribution Chart

The categorical distributions shown in Figure 3 indicate a well-balanced gender distribution and a customer base that is mainly composed of college graduates, followed by bachelor and high school graduates, with master's degree holders being the smallest group. The regional distribution is fairly evenly distributed, but the East and West regions are slightly overrepresented in comparison to the North and South regions. The biggest group of customers are Regular loyalty members and Rare buyers, so the buying frequency is quite low in general. The most purchased product category is Electronics, and the second is Clothing, while Home and Beauty has the lowest purchase shares.

4) Correlation Analysis

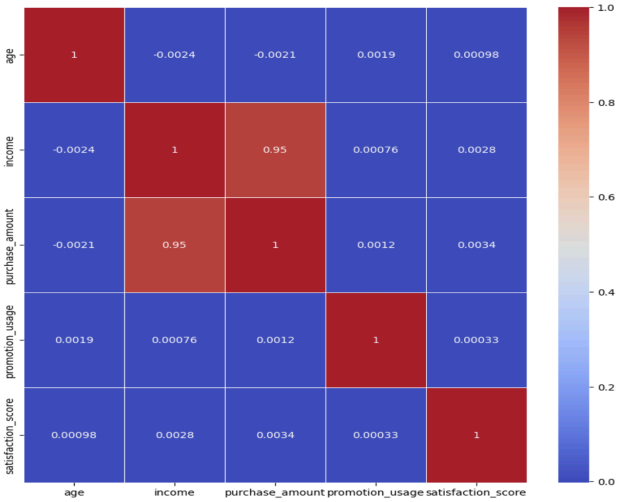


Fig. 4: Heatmap Correlation

The correlation analysis in figure 4 indicates that the highest correlation is between income and purchase_amount (r = 0.95). This means that customers with higher income tend to purchase more. In contrast, the other numerical variables are only weakly correlated, which indicates that they are mostly independent. That is, the dataset provides little evidence of multicollinearity, which is a good sign for regression-based modeling.

IV. Data Analysis and Modelling

A. *Preprocessing Before Modeling*

Prior to model training, three preprocessing steps were performed. First, categorical variables (gender, education, region, loyalty_status, purchase_frequency and product_category) were transformed to numerical values using Label Encoding. Then, the dataset was split into training and testing sets with an 80:20 ratio with random_state=42, leading to 79,586 training samples and 19,897 testing samples. Finally, the numeric features were normalized by using StandardScaler, which is very important for the SGD Regressor since it is sensitive to feature scale differences [21].

B. *Models Used*

To predict customer income three regression models were used. Linear Regression: Linear regression was the baseline model assuming a linear relationship between features and the target variable. Its performance sets a benchmark for more complex models [22].

This project constructed and evaluated three machine learning models:

1) Linear Regression

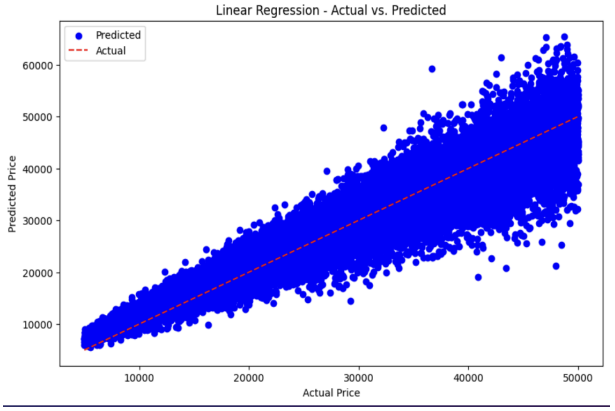


Fig. 5: Actual vs Predicted Income Linear Regression

Linear Regression was used as a baseline model to predict the target variable assuming linear relationship between input features and target variable. We trained the model using the closed-form solution of Ordinary Least Squares (OLS) and hence we do not need to train the model iteratively or for epochs. The prediction results are shown in Figure 5. It can be seen that the model performs reasonably well for lower income values, where the predicted values are close to the actual values. Prediction errors, however, become more pronounced as income rises. The larger dispersion of prediction points in higher income ranges indicates that the model cannot learn more complex patterns in the data and so is less reliable for predicting larger values of income.

2) Random Forest

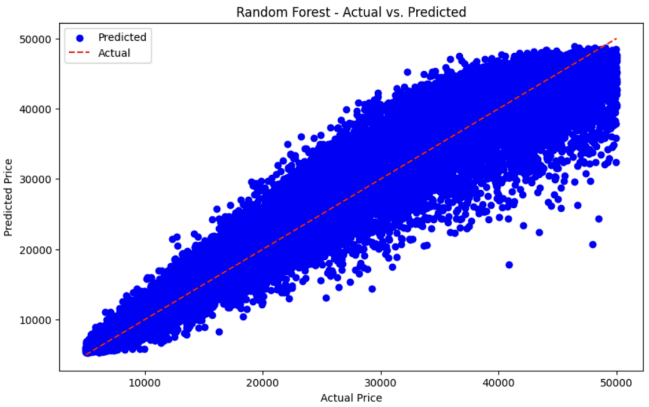


Fig. 6: Actual vs Predicted Income Random Forest

Random Forest is an ensemble learning process that makes decision trees at the same time and then puts their predictions together to get the final answer. It is different from Linear Regression because it can understand relationships in the data that are not straight lines. This helps to reduce the chance of the model being too perfect for the data it was trained on.

At figure 6, it can be seen that the predictions made by Random Forest are really close to the values for most income ranges. The mistakes in the predictions are also pretty steady for people with medium to high incomes.. There are not as many really bad guesses. This means that Random Forest is good at finding the patterns in the data and making predictions that are trusted. Random Forest is better at modeling the data and giving predictions.

3) SGD Regressor

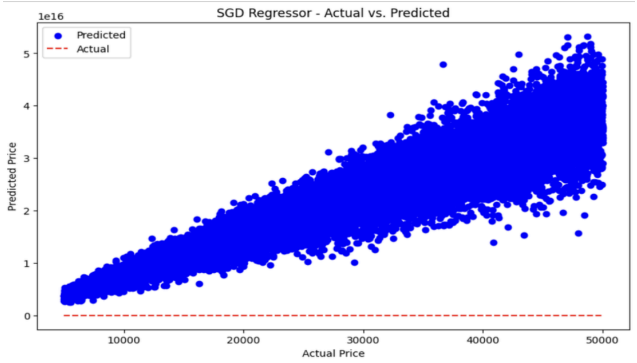


Fig. 7: Actual vs Predicted Income Random Forest

Stochastic Gradient Descent SGD Regressor is a linear model that solves the loss function in batch. It is computationally efficient and suitable for large data sets but its performance is highly dependent on the correct feature scaling and hyperparameter tuning. The model in this study predicted extremely large values, many orders of magnitude larger than actual income. Some predictions were on the order of 1016 while actual values were in the range of 0 to 50,000. Consequently, the actual values became almost indistinguishable from the predicted outputs as shown in figure 7. These results suggest that the model has not learned any meaningful patterns from the data in its current configuration. The poor performance demonstrates the sensitivity of SGD to data scaling and parameter settings, indicating that further tuning is necessary before the model can produce reliable predictions.

C. Model Evaluation Results (Before Tuning)

Three machine learning models were trained and tested on the test dataset. The model performance was evaluated based on Accuracy (calculated from MAPE as 10−MAPE×100100−MAPE×100), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R2 Score. The evaluation results are summarized in Fig 8.

	Model	Epochs	Accuracy (%)	MAE	MSE	RMSE	R2
0	Linear Regression	N/A (Closed-form)	8.694000e+01	3.097530e+03	1.696655e+07	4.119050e+03	8.988000e-01
1	Random Forest	N/A (Closed-form)	8.864000e+01	2.873230e+03	1.477572e+07	3.843920e+03	9.119000e-01
2	SGD Regressor	407	-7.641167e+13	2.092280e+16	5.467596e+32	2.338289e+16	-3.260043e+24

Fig. 8: Model Comparison Before Tuning

The evaluation results shown in figure 8 show that Linear Regression and Random Forest were able to make good predictions on customer income, while SGD Regressor was not able to make reliable predictions in its first configuration. Linear Regression got an accuracy of 86.94% and a R² Score of 0.8988, which means that it could explain almost 90% of the variance in the target variable. However, the prediction errors increased with higher income levels suggesting that the model was less successful in capturing more complex relationships in the data.

The Random Forest gave the overall best performance with an accuracy of 88.64%, MAE of 2,873.23, RMSE of 3,843.92 and an R2 Score of 0.9119. The model explained around 91.19% of variability in income and predicted values were more consistently closer to the actual values with fewer extreme errors than Linear Regression.

In contrast, the initial SGD Regressor model produced extremely poor results, reflected by a highly negative R² Score and unrealistic prediction values. This behavior was mainly caused by the model's sensitivity to feature scaling[24]. The results highlight the importance of applying data normalization before training SGD-based models.

D. Hyparparameter Tuning

Further to optimize the model performance hyper-parameter tuning was done by using GridSearchCV with 3-fold cross validation for both Random Forest and SGD Regressor. The search space for Random Forest was n_estimators (50, 100, 200), max_depth (None, 10, 20) and min_samples_split (2, 5). For SGD Regressor, the parameters explored were max_iter (500, 1000, 2000), alpha (0.0001, 0.001, 0.01) and learning_rate ('constant', 'optimal', 'adaptive').

	Model	Accuracy (%)	R2
0	Random Forest (Before)	8.864000e+01	9.119000e-01
1	Random Forest (Tuned)	8.913000e+01	9.176000e-01
2	SGD Regressor (Before)	-7.641167e+13	-3.260043e+24
3	SGD Regressor (Tuned)	8.693000e+01	8.988000e-01

Fig. 9: Comparison before and after tuning

The tuning process increased the performance of both models. With Random Forest, prediction errors were lower and R2 Score was high and significant improvement was noted with SGD Regressor over its untuned version. Figure 9 presents the comparison of model performance before and after tuning.

E. Relative Error Test on Sample Predictions

The predictions of the Random Forest were tested against 5 randomly sampled actual values (random_state=42) to concretely validate the model performance. The results showed low relative error in all samples, implying that the model consistently produces predictions near the true values.

V. Interpretation of Results

A. Best Performing Model

The tuned Random Forest Regressor performed best with a R² Score of 0.9176, an accuracy of 89.13%, an MAE of 2,772.27 and an RMSE of 3,717.20 based on the overall evaluation results. The model could capture non-linear relationships in the data better than Linear Regression. It performed better than the SGD Regressor, which was more sensitive to feature scaling and model configuration. The results indicate that Random Forest is the best model for predicting customer income in this study.

B. Factors Influencing Income

The correlation heatmap shows the purchase_amount is the most important variable with very strong positive

correlation with income (r = 0.95) [25]. This result indicates that higher income groups of customers are generally associated with higher purchase amount. On the other hand, variables such as age, promotion_usage and satisfaction_score have correlation coefficients close to 0, which means that they are weakly or not linearly related to income. These results show that purchase behavior is a much better predictor of customer income than demographic features or any satisfaction-related factors in this dataset.

VI. Conclusion

This project shows how to predict customers income using EDA and Machine Learning techniques with the demographic data of customers and their purchasing behavior. Based on the EDA results, we found that most features were relatively normally distributed. We also found that income was most related to purchase amount. From our correlation heatmap we could see the more the customers' income the more they tend to spend. Therefore purchase amount was a major predictor of income. The correlation coefficient was 0.95 which confirmed the strong relation. The purchase amount was the most influential predictor in the dataset.

Our models were trained on linear regression, random forest regressor and SGD regressor. From our prediction results, we found Random Forest had the best performance. Random Forest was able to learn more complex patterns in the data and gave us predictions closer to the actual income. After tuning the hyperparameters, the tuned Random Forest Regressor gave the best performance with an accuracy of 89.13% and R^2 Score of 0.9176. Linear Regression was able to be our baseline model but did not perform as well for higher income customers. The SGD Regressor showed lower accuracy and higher variance. The results of SGD Regressor also pointed out the importance of feature scaling and hyperparameter tuning for gradient-based learning methods.

The conclusion from this project is that the choice of a machine learning model is critical to generate accurate predictions. The clean dataset without missing values and duplicate records improved the quality and reliability of the models to a great deal. The data and insights collected through this project can be used for customer segmentation, marketing strategies and business decision making. In the future, more features and more advanced machine learning models could be explored to improve the prediction performance and operational efficiency.

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